

- **Module-4 (8 hours)**

- Embedded Systems – Definition, Embedded systems vs general computing systems,
- Classification of Embedded Systems,
- Major application areas of Embedded Systems,
- Elements of an Embedded System,
- Core of the Embedded System,
- Microprocessor vs Microcontroller, RISC vs CISC,
- Sensors and Interfacing – Instrumentation and control systems,
- Transducers, Sensors, Actuators,
- LED, 7-Segment LED Display.

*SUNIL .M
ASSISTANT PROFESSOR,
ELECTRONICS AND COMMUNICATION ENGG.
JAIN INSTITUTE OF TECHNOLOGY,
DAVANGERE*

What is Embedded System?

- An Electronic/Electro mechanical system which is designed to perform a specific function and is a combination of both hardware and firmware (Software).
- E.g. Electronic Toys, Mobile Handsets, Washing Machines, Air Conditioners, Automotive Control Units, Set Top Box, DVD Player etc...
- Embedded Systems are:
 - ☐ Unique in character and behaviour.
 - ☐ With specialized hardware and software.

Embedded Systems Vs General Computing Systems:

General Purpose Computing System	Embedded System
A system which is a combination of generic hardware and General Purpose Operating System for executing a variety of applications.	A system which is a combination of special purpose hardware and embedded OS for executing a specific set of applications.
Contain a General Purpose Operating System (GPOS)	May or may not contain an operating system for functioning
Applications are alterable (programmable) by user (It is possible for the end user to re-install the Operating System, and add or remove user applications)	The firmware of the embedded system is pre-programmed and it is non-alterable by end-user
Performance is the key deciding factor on the selection of the system. Always „Faster is Better“	Application specific requirements (like performance, power requirements, memory usage etc) are the key deciding factors
Less/not at all tailored towards reduced operating power requirements, options for different levels of power management.	Highly tailored to take advantage of the power saving modes supported by hardware and Operating System
Response requirements are not time critical	For certain category of embedded systems like mission critical systems, the response time requirement is highly critical
Need not be deterministic in execution behaviour	Execution behaviour is deterministic for certain type of embedded systems like „Hard Real Time“ systems

Classification of Embedded Systems:

- ☐ Based on Generation
- ☐ Based on Complexity & Performance Requirements

- ☐ Based on deterministic behavior
- ☐ Based on Triggering

1. Classification based on Generation

First Generation: The early embedded systems built around 8-bit microprocessors like 8085 and Z80 and 4-bit microcontrollers.

EX. stepper motor control units, Digital Telephone Keypads etc.

Second Generation: Embedded Systems built around 16-bit microprocessors and 8 or 16-bit microcontrollers, following the first generation embedded systems.

EX. SCADA, Data Acquisition Systems etc.

Third Generation: Embedded Systems built around high performance 16/32 bit Microprocessors/controllers, Application Specific Instruction set processors like Digital Signal Processors (DSPs), and Application Specific Integrated Circuits (ASICs). The instruction set is complex and powerful.

EX. Robotics, industrial process control, networking etc.

Fourth Generation: Embedded Systems built around System on Chips (SoC's), Re-configurable processors and multicore processors. It brings high performance, tight integration and miniaturization into the embedded device market.

EX Smart phone devices, MIDs etc.

2. Classification based on Complexity & Performance.

➤ **Small Scale:** The embedded systems built around low performance and low cost 8 or 16 bit microprocessors/ microcontrollers. It is suitable for simple applications and where performance is not time critical. It may or may not contain OS.

➤ **Medium Scale:** Embedded Systems built around medium performance, low cost 16 or 32 bit microprocessors / microcontrollers or DSPs. These are slightly complex in hardware and firmware. It may contain GPOS/RTOS.

➤ **Large Scale/Complex:** Embedded Systems built around high performance 32 or 64 bit RISC processors/controllers, RSoC or multi-core processors and PLD. It requires

complex hardware and software. These system may contain multiple processors/controllers and co-units/hardware accelerators for offloading the processing requirements from the main processor. It contains RTOS for scheduling, prioritization and management.

3 Classification Based on deterministic behaviour:

It is applicable for Real Time systems. The application/task execution behaviour for an embedded system can be either deterministic or non-deterministic.

These are classified in to two types.

1. **Soft Real time Systems:** Missing a deadline may not be critical and can be tolerated to a certain degree
2. **Hard Real time systems:** Missing a program/task execution time deadline can have catastrophic consequences (financial, human loss of life, etc.)

4. Classification Based on Triggering:

These are classified into two types.

- **Event Triggered:** Activities within the system (e.g., task run-times) are dynamic and depend upon occurrence of different events.
- **Time triggered:** Activities within the system follow a statically computed schedule (i.e., they are allocated time slots during which they can take place) and thus by nature are predictable.

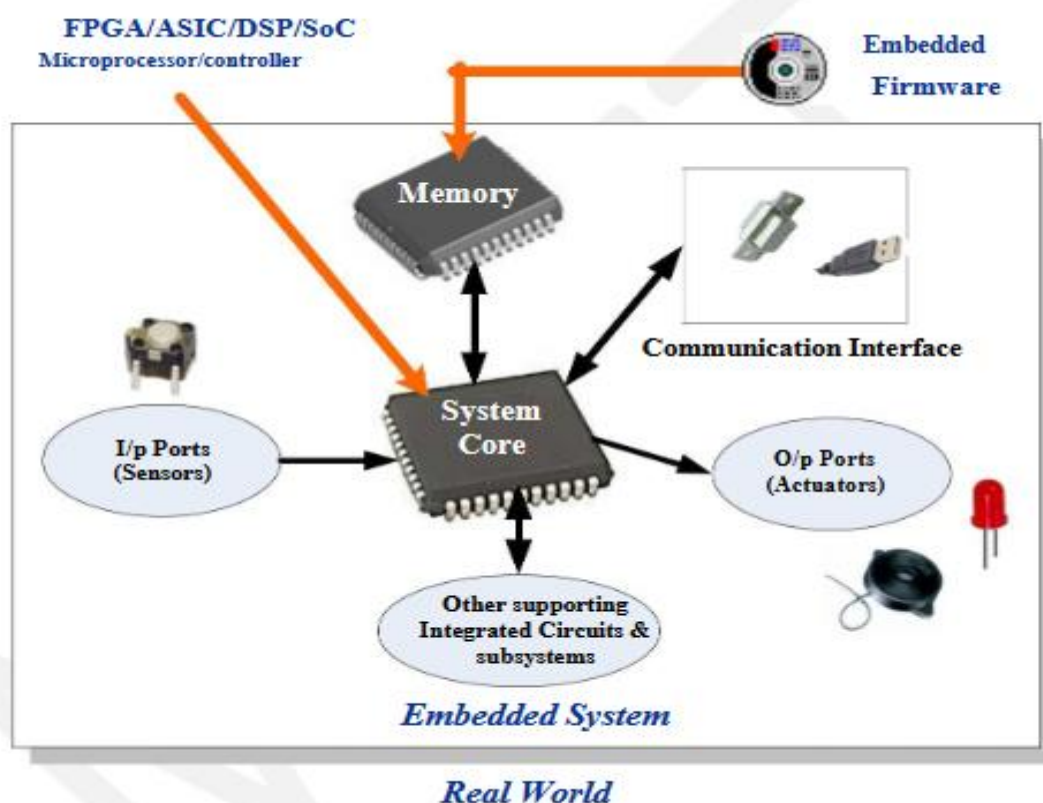
Major Application Areas of Embedded Systems:

- **Consumer Electronics:** Camcorders, Cameras etc.
- **Household Appliances:** Television, DVD players, washing machine, Fridge, Microwave Oven etc.
- **Home Automation and Security Systems:** Air conditioners, sprinklers, Intruder detection alarms, Closed Circuit Television Cameras, Fire alarms etc.

- **Automotive Industry:** Anti-lock breaking systems (ABS), Engine Control, Ignition Systems, Automatic Navigation Systems etc.
- **Telecom:** Cellular Telephones, Telephone switches, Handset Multimedia Applications etc.
- **Computer Peripherals:** Printers, Scanners, Fax machines etc.
- **Computer Networking Systems:** Network Routers, Switches, Hubs, Firewalls etc.
- **Health Care:** Different Kinds of Scanners, EEG, ECG Machines etc.
- **Measurement & Instrumentation:** Digital multi meters, Digital CROs, Logic Analyzers PLCsystems etc.
- **Banking & Retail:** Automatic Teller Machines (ATM) and Currency counters, Point of Sales (POS).
- **Card Readers:** Barcode, Smart Card Readers, Hand held Devices etc.

- **ELEMENTS OF EMBEDDED SYSTEMS:**

- An embedded system is a combination of 3 things, Hardware Software Mechanical Components and it is supposed to do one specific task only. A typical embedded system contains a single chip controller which acts as the master brain of the system. Diagrammatically an embedded system can be represented as follows:



Embedded systems are basically designed to regulate a physical variable (such as Microwave Oven) or to manipulate the state of some devices by sending some signals to the actuators or devices connected to the output port system (such as temperature in Air Conditioner), in response to the input signal provided by the end users or sensors which are connected to the input ports. Hence the embedded systems can be viewed as a reactive system.

The control is achieved by processing the information coming from the sensors and user interfaces and controlling some actuators that regulate the physical variable. Keyboards, push button, switches, etc. are Examples of common user interface input devices and LEDs, LCDs, Piezoelectric buzzers, etc examples for common user interface output devices for a typical embedded system. The requirement of type of user interface changes from application to application based on domain.

The memory of the system is responsible for holding the code (control algorithm and other important configuration details). There are two types of memories are used in any embedded system. Fixed memory (ROM) is used for storing code or program. The user cannot change the firmware in this type of memory. The most common types of memories used in embedded systems for control algorithm storage are OTP, PROM, UVEPROM, EEPROM and FLASH.

Memory for implementing the code may be present on the processor or may be implemented as a separate chip interfacing the processor. In a controller based embedded system, the controller may contain internal memory for storing code such controllers are called Micro-controllers with on-chip ROM, eg. Atmel AT89C51.

The Core of the Embedded Systems:

The core of the embedded system falls into any one of the following categories.

- **General Purpose and Domain Specific Processors**
 - Microprocessors,
 - Microcontrollers,

- Digital Signal Processors.
- **Programmable Logic Devices (PLDs)**
- **Application Specific Integrated Circuits (ASICs)**
- **Commercial off the shelf Components (COTS)**

GENERAL PURPOSE AND DOMAIN SPECIFIC PROCESSOR:

Almost 80% of the embedded systems are processor/ controller based. The processor may be microprocessor or a microcontroller or digital signal processor, depending on the domain and application.

Microprocessor:

A silicon chip representing a Central Processing Unit (CPU), which is capable of performing arithmetic as well as logical operations according to a pre-defined set of Instructions, which is specific to the manufacturer.

In general the CPU contains the Arithmetic and Logic Unit (ALU), Control Unit and Working registers. Microprocessor is a dependant unit and it requires the combination of other hardware like Memory, Timer Unit, and Interrupt Controller etc for proper functioning. Intel claims the credit for developing the first Microprocessor unit Intel 4004, a 4 bit processor which was released in Nov 1971. Developers of microprocessors.

Intel – Intel 4004 – November 1971(4-bit)

Intel – Intel 4040.

Intel – Intel 8008 – April 1972.

Intel – Intel 8080 – April 1974(8-bit).

Motorola – Motorola 6800.

Intel – Intel 8085 – 1976.

Zilog - Z80 – July 1976.

Microcontroller:

- A highly integrated silicon chip containing a CPU, scratch pad RAM, Special and General purpose Register Arrays, On Chip ROM/FLASH memory for program storage, Timer and Interrupt control units and dedicated I/O ports .
- Microcontrollers can be considered as a super set of Microprocessors.
- Microcontroller can be general purpose (like Intel 8051, designed for generic applications and domains) or application specific (Like Automotive AVR from Atmel Corporation. Designed specifically for automotive applications).
- Since a microcontroller contains all the necessary functional blocks for independent working, they found greater place in the embedded domain in place of microprocessors .
- Microcontrollers are cheap, cost effective and are readily available in the market.
- Texas Instruments TMS 1000 is considered as the world's first microcontroller.

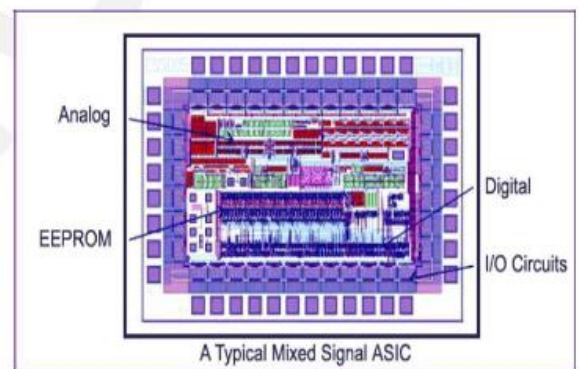
Digital Signal Processors (DSPs):

- Powerful special purpose 8/16/32 bit microprocessors designed specifically to meet the computational demands and power constraints of today's embedded audio, video, and communications applications.
- Digital Signal Processors are 2 to 3 times faster than the general purpose microprocessors in signal processing applications.
- DSPs implement algorithms in hardware which speeds up the execution whereas general purpose processors implement the algorithm in firmware and the speed of execution depends primarily on the clock for the processors.
- DSP can be viewed as a microchip designed for performing high speed computational operations for „addition“, „subtraction“, „multiplication“ and „division“.
- A typical Digital Signal Processor incorporates the following key units
 - Program Memory
 - Data Memory
 - Computational Engine
 - I/O Unit

Audio video signal processing, telecommunication and multimedia applications are typical examples where DSP is employed.

Application Specific Integrated Circuit (ASIC):

- A microchip designed to perform a specific or unique application. It is used as replacement to conventional general purpose logic chips.
- ASIC integrates several functions into a single chip and thereby reduces the system development cost.
- Most of the ASICs are proprietary products. As a single chip, ASIC consumes very small area in the total system and thereby helps in the design of smaller systems with high capabilities/functionalities.
- ASICs can be pre-fabricated for a special application or it can be custom fabricated by using the components from a re-usable „building block“ library of components for a particular customer application.
- Fabrication of ASICs requires a non-refundable initial investment (Non Recurring Engineering (NRE) charges) for the process technology and configuration expenses



Programmable Logic Devices (PLDs):

- Logic devices provide specific functions, including device-to-device interfacing, data communication, signal processing, data display, timing and control operations, and almost every other function a system must perform.
- Programmable logic devices (PLDs) offer customers a wide range of logic capacity, features, speed, and voltage characteristics - and these devices can be re-configured to perform any number of functions at any time.

- Designers can use inexpensive software tools to quickly develop, simulate, and test their logic designs in PLD based design. The design can be quickly programmed into a device, and immediately tested in a live circuit.
- PLDs are based on re-writable memory technology and the device is reprogrammed to change the design.
- Field Programmable Gate Arrays (FPGAs) and Complex Programmable Logic Devices (CPLDs) are the two major types of programmable logic devices
- **FPGA:**

FPGA is an IC designed to be configured by a designer after manufacturing. FPGAs offer the highest amount of logic density, the most features, and the highest performance. Logic gate is Medium to high density ranging from 1K to 500K system gates.
- **CPLD:**

CPLDs offer very predictable timing characteristics and are therefore ideal for critical control applications.
- CPLDs such as the Xilinx Cool Runner series also require extremely low amounts of power and are very inexpensive, making them ideal for cost-sensitive, battery-operated, portable applications such as mobile phones and digital handheld assistants.

Commercial off the Shelf Component (COTS):

- A Commercial off-the-shelf (COTS) product are designed in such a way to provide easy integration and interoperability with existing system components.
- Typical examples for the COTS hardware unit are Remote Controlled Toy Car control unit including the RF Circuitry part, High performance, high frequency microwave electronics (2 to 200 GHz), High bandwidth analog-to-digital converters, Devices and components for operation at very high temperatures, Electro-optic IR imaging arrays, UV/IR Detectors etc.
- A COTS component in turn contains a General Purpose Processor (GPP) or Application Specific Instruction Set Processor (ASIP) or Application Specific Integrated Chip(ASIC)/Application Specific Standard Product (ASSP) or Programmable Logic Device (PLD).

Microprocessor Vs Microcontroller:

Microprocessor	Microcontroller
A silicon chip representing a Central Processing Unit (CPU) which is capable of performing arithmetic as well as logical operations according to a pre-defined set of instruction.	A microcontroller is a highly integrated chip that, contains a CPU, scratch pad RAM, Special and General purpose Register Arrays, On Chip of Instructions ROM/FLASH memory for program storage, Timer and Interrupt control units and dedicated I/O ports.
It is a dependent unit. It requires the combination of other chips like Timers, Program and data memory chips, Interrupt controllers etc for functioning	It is a self-contained unit and it doesn't require external Interrupt Controller, Timer, UART etc for its functioning.
Most of the time general purpose in design and operation	Mostly application oriented or domain specific
Doesn't contain a built in I/O port. The I/O Port functionality needs to be implemented with the help of external Programmable Peripheral Interface Chips like 8255	Most of the processors contain multiple built in I/O ports which can be operated as a single 8 or 16 or 32 bit Port or as individual port pins.
Targeted for high end market where performance is important.	Targeted for embedded market where performance is not so critical (At present this demarcation is invalid)
Limited power saving options compared to	Includes lot of power saving features

RISC V/s CISC Processors/Controllers:

RISC	CISC
Lesser no. of instructions	Greater no. of Instructions
Instruction Pipelining and increased execution speed	Generally no instruction pipelining feature
Orthogonal Instruction Set (Allows each instruction to operate on any register and use any addressing mode)	Non Orthogonal Instruction Set (All instructions are not allowed to operate on any register and use any addressing mode. It is instruction specific)
Operations are performed on registers only, the only memory operations are load and store	Operations are performed on registers or memory depending on the instruction
Large number of registers are available	Limited no. of general purpose registers
Programmer needs to write more code to execute a task since the instructions are simpler ones.	. A programmer can achieve the desired functionality with a single instruction which in turn provides the effect of using more simpler single instructions in RISC.
Single, Fixed length Instructions.	Variable length Instructions.
Less Silicon usage and pin count.	More silicon usage since more additional decoder logic is required to implement the complex instruction decoding.
With Harvard Architecture	Can be Harvard or Von- Neumann Architecture.

Sensors & Actuators:

- Embedded system is in constant interaction with the real world.
- Controlling/monitoring functions executed by the embedded system is achieved in accordance with the changes happening to the Real World.
- The changes in the system environment or variables are detected by the sensors connected to the input port of the embedded system.
- If the embedded system is designed for any controlling purpose, the system will produce some changes in controlling variable to bring the controlled variable to the desired value.
- It is achieved through an actuator connected to the out port of the embedded system.

➤ **Sensor:**

A transducer device which converts energy from one form to another for any measurement or control purpose. Sensors acts as input device.

- Eg. Hall Effect Sensor which measures the distance between the cushion and magnet in the Smart Running shoes from Adidas.
- Example: IR, humidity , PIR(passive infra-red) , ultrasonic , piezoelectric , smoke sensors.



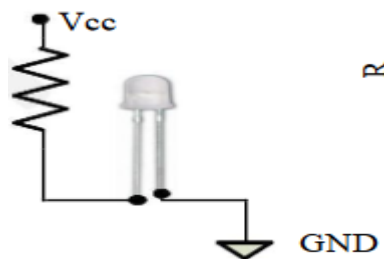
Actuator:

A form of transducer device (mechanical or electrical) which converts signals to corresponding physical action (motion). Actuator acts as an output device. Eg. Micro motor actuator which adjusts the position of the cushioning element in the Smart Running shoes from Adidas.



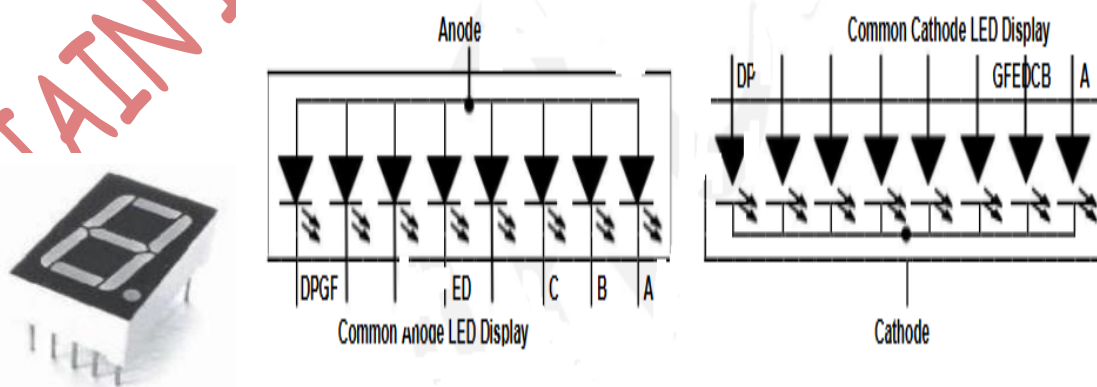
Light Emitting Diode (LED):

- Light Emitting Diode (LED) is an output device for visual indication in any embedded system
- LED can be used as an indicator for the status of various signals or situations.
- Typical examples are indicating the presence of power conditions like „Device ON“, „Battery low“ or „Charging of battery“ for a battery operated handheld embedded devices.
- LED is a p-n junction diode and it contains an anode and a cathode.
- For proper functioning of the LED, the anode of it should be connected to +ve terminal of the supply voltage and cathode to the –ve terminal of supply voltage.
- The current flowing through the LED must limited to a value below the maximum current that it can conduct.
- A resistor is used in series between the power supply and the resistor to limit the current through the LED.



➤ 7-Segment LED Display

- The 7 – segment LED display is an output device for displaying alpha numeric characters.
- It contains 8 light-emitting diode (LED) segments arranged in a special form. Out of the 8 LED segments, 7 are used for displaying alpha numeric characters.
- The LED segments are named A to G and the decimal point LED segment is named as DP.
- The LED Segments A to G and DP should be lit accordingly to display numbers and characters.
- The 7 – segment LED displays are available in two different configurations, namely; Common anode and Common cathode.
- In the Common anode configuration, the anodes of the 8 segments are connected commonly whereas in the Common cathode configuration, the 8 LED segments share a common cathode line.
- Based on the configuration of the 7 – segment LED unit, the LED segment anode or cathode is connected to the Port of the processor/controller in the order „A“ segment to the least significant port Pin and DP segment to the most significant Port Pin.
- The current flow through each of the LED segments should be limited to the maximum value supported by the LED display unit.
- The typical value for the current falls within the range of 20Ma.
- The current through each segment can be limited by connecting a current limiting resistor to the anode or cathode of each segment.



Transducer

A **transducer** is a device that converts energy from one form to another.

Usually a transducer converts a signal in one form of energy to a signal in another.

Transducers are often employed at the boundaries of automation, measurement, and control systems, where electrical signals are converted to and from other physical quantities (energy, force, torque, light, motion, position, etc.).

The process of converting one form of energy to another is known as transduction.

Transducers can be categorized by which direction information passes through them:

Sensors and actuators are the type transducers.

Applications of transducers are:

- Cathode-ray tubes (CRT) – converts electrical signals into visual signals.
- Loudspeakers, earphones – converts electrical signals into sound.
- Microphones – converts sound into an electrical signal.
- Tape head, disk read-and-write heads – converts magnetic fields on a magnetic medium to and from electrical signals
- Antennae – converts propagating electromagnetic waves to and from conducted electrical signals.



INSRUMENTATION AND CONTROL SYSTEMS

- An instrumentation system obtains data about a physical system either for the purpose of collecting information about that physical system or for the feedback control of the physical system.
- Any instrumentation system must include an input transducer (sensor), whose response to a particular stimulus can be measured electrically.
- The other component that is generally present in modern instrumentation systems is a digital processor, such as a computer or a micro-controller.
- These programmable components have the flexibility to be used for a variety of functions.
- The most important function that they perform is to convert data into information.

- In the simplest situation the processing required to extract information may only involve converting an input signal by a scale factor so that the final result is in conventional units.
- In the electronics industry, **embedded instrumentation** refers to the integration of test and measurement instrumentation into semiconductor chips (or integrated circuit devices).
- Embedded instrumentation differs from embedded system, which are electronic systems or subsystems that usually comprise the control portion of a larger electronic system.
- Instrumentation embedded into chips (embedded instrumentation) is employed in a variety of electronic test applications, including validating and testing chips themselves, validating, testing and debugging the circuit boards where these chips are deployed, and troubleshooting systems once they have been installed in the field.